

Quantile Regression with Censored Selection and Many Controls

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Outline

- 1 Introduction
- 2 Estimator
- 3 Asymptotic Results (Technical parts)
- 4 Simulation
- 5 An application: Wage Inequality in the United States

Motivation

Core Issue: Non-Random Sample Selection + Censored Data

- Individuals **self-select** into groups based on unobserved characteristics
- Income survey data are often **censored** (e.g., top-coding or hours worked bounded by zero)

Example: Wage Analysis

- Only observe wages for **employed** individuals → Selection
- Hours worked: $Y_2 = \max\{Y_2^*, 0\}$ → Censoring



Both issues distort wage distribution

Empirical Illustration



Data

American Community Survey
2001–2023

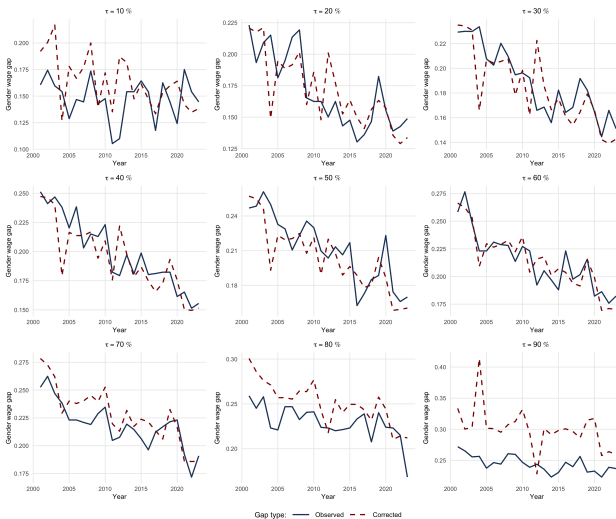
Method

Selection-corrected
wage predictions

**Goal: Correct for both selection bias &
censoring simultaneously**

Next: results for gender wage gaps across quantiles

Key Findings



Gender wage gaps across quantiles (2001 - 2023)

Literature Review

I. Sample Selection

- **Mean regression methods**

Heckman (1974, 1979), Ahn and Powell (1993), Andrews and Schafgans (1998), Chen and Khan (2003), Das et al. (2003)

- **Quantile regression methods**

- **Control function approach:** Buchinsky and Hahn (1998), Buchinsky (2001)
- **Copula function approach (method-of-moments)**
 - Arellano and Bonhomme (2017)
 - Chen and Wang (2023): Censoring
 - Chen et al. (2024): [Censored selection](#)

II. Censored Quantile Regression (CQR)

- **Non-convex optimization methods**

Powell (1986), Buchinsky (1994), Fitzenberger (1997), Fitzenberger and Winker (2007), Koenker (2008)

- **Subsample methods**

- **Recursive:** Portnoy (2003), Peng & Huang (2008)
- **Sequential:** [Chen \(2018\)](#)

Our Work

I. Sample Selection

- **Mean regression methods**
Heckman (1974, 1979), Ahn and Powell (1993), Andrews and Schafgans (1998), Chen and Khan (2003), Das et al. (2003)
 - **Quantile regression methods**
 - **Control function approach:** Buchinsky and Hahn (1998), Buchinsky (2001)
 - **Copula function approach (method-of-moments)**
 - Arellano and Bonhomme (2017)
 - Chen and Wang (2023): Censoring
 - Chen et al. (2024): [Censored selection](#)
- + **Copula function approach (nonlinear least square)**

II. Censored Quantile Regression (CQR)

- **Non-convex optimization methods**
Powell (1986), Buchinsky (1994), Fitzenberger (1997), Fitzenberger and Winker (2007), Koenker (2008)
- **Subsample methods**
 - Recursive: Portnoy (2003), Peng & Huang (2008)
 - Sequential: [Chen \(2018\)](#)

+

III. Machine Learning Methods

- Copula identification via Nonlinear Least Squares (NLS)
- Censored selection provides more information
- ℓ_1 -penalized QR + Hard-thresholding

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Model

$$\begin{aligned}Y_1^* &= X' \beta_0(U), \\Y_2 &= \max\{Y_2^*, 0\} = \max\{X' \gamma_0(V), 0\}, \\D &= 1\{Y_2^* > 0\}, \\Y_1 &= DY_1^*\end{aligned}\tag{1}$$

- Y_1^* and Y_2^* are log-wage level and hours worked
- D indicates labor force participation
- X are determinants of both levels, such as working experience, education, number of children and, etc

Correcting for Selection Bias

Identification Challenge

- Model 1 depends on two sources of unobserved heterogeneity:
 - Latent outcome rank U
 - Percentile V
- **Bias source:** Dependence between U and V causes sample selection bias
- **Solution:** Control the dependence structure (**copula functions**)

Identification

Selection Bias Correction

$$\begin{aligned} & P(Y_1 < X'\beta_0(u) | Y_2^* > X'\gamma_0(v), X = x) \\ &= P(Y_1^* < X'\beta_0(u) | X'\gamma_0(V) > X'\gamma_0(v), X = x) \\ &= \frac{C(u, 1, \rho_0) - C(u, v, \rho_0)}{1 - v} \\ &\equiv G(u, v, \rho_0), \end{aligned} \tag{2}$$

Given any pair of $u, v \in \mathcal{U}$ and for observations with $X'\gamma(v) > 0$

Interpretation

- $C(\cdot)$: Copula function capturing **dependence**
- ρ_0 : Dependence parameter
- $G(u, v, \rho_0)$: Identified function for estimation

Identification

Conditional moments

- Identification via conditional moment:
 - Method-of-Moment
 $E(\varepsilon|X) = 0$
- Estimation by its sample average
 - $\min \frac{1}{n} \sum_{i=1}^n \varepsilon_i \cdot g(X_i)$
 - Requires choosing instruments $g(X)$
 - Limited information:
 - $X, X^2, p(X)$, etc
 - Arbitrary selection
 - Inefficient: Uses only selected moments

Unconditional moments

- Identification via unconditional moments:
 - Nonlinear Least Square
 $\min E(Y - m(X))^2$
- Estimation by its sample average
 - $\min \frac{1}{n} \sum_{i=1}^n (Y_i - m(X_i))^2$
 - No instrument selection
 - Full information:
 - No arbitrary choices
 - Efficient: Uses infinite moments

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Identification

Identification via unconditional moments

By the monotonicity of the cumulative distribution function, let

$$u = G^{-1}(\tau, v, \rho_0), \quad (3)$$

$$\begin{aligned} & E[1\{Y_1 < X'\beta_0(u)\} | X, Y_2 > X'\gamma_0(\tilde{v}) > 0] \\ &= \frac{P(U < G^{-1}(\tau_m, v, \rho_0), V > \tilde{v} | X)}{P(V > \tilde{v} | X)} \\ &= \frac{C(G^{-1}(\tau_m, v, \rho_0), 1, \rho_0) - C(G^{-1}(\tau_m, v, \rho_0), \tilde{v}, \rho_0)}{1 - \tilde{v}} \\ &= G(G^{-1}(\tau_m, v, \rho_0), \tilde{v}, \rho_0), \end{aligned} \quad (4)$$

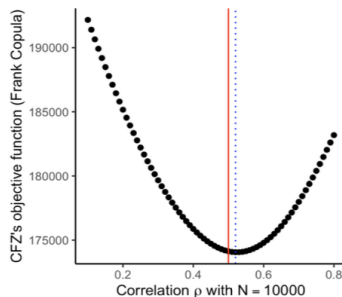
with $X'\gamma(\tilde{v}) > 0$ by choosing some $v, \tilde{v}$ and a finite grid $\tau_1 < \tau_2 < \dots < \tau_M$ on $\mathcal{U}$.

Estimation

Final Step: estimate the copula parameter ρ_0

we get $\hat{\rho}$ via

$$\min_{\rho \in \mathbb{R}} \int_{\tilde{v}} \int_{\nu} \sum_{m=1}^M \sum_{i=1}^n \left[1\{Y_{1i} < X_i' \hat{b}(\tau_m, \nu)\} - G^*(\tau_m, \nu, \tilde{v}, \rho) \right]^2 \\ 1\{Y_{2i} > X_i' \hat{\gamma}(\tilde{v}) > 0\} d\nu d\tilde{v}.$$



Estimation

First Step (selection equation): estimate parameters $\gamma_0(v)$

In model 1, we have

$$Y_2 = \max\{Y_2^*, 0\} = \max\{X' \gamma_0(V), 0\}$$

$$D = 1\{Y_2^* > 0\}$$

Due to the non-convex nature of the optimization problem in censored quantile regression, our model is based on Chen's (JOE, 2018). Define $\tilde{\gamma}(v_0)$ as the solution to

$$\min_{\gamma \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n \rho_{v_0}(Y_{2i} - X_i' \gamma) + \frac{\lambda}{n} \|\gamma\|_{1,n}, \quad (5)$$

and $\tilde{\gamma}(v_{l+1})$ as the solution to

$$\min_{\gamma \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n 1\{X_i' \tilde{\gamma}(v_l) > \delta_n\} \rho_{v_{l+1}}(Y_{2i} - X_i' \gamma) + \frac{\lambda}{n} \|\gamma\|_{1,n}, \quad (6)$$

for $v_{L_n} < \dots < v_1 < v_0 \in \mathcal{U}$.

Estimation

First Step (selection equation): estimate parameters $\gamma_0(v)$

Addressing Convergence Rate and Estimation Bias

- **Sparsity**: Only a small subset of variables truly matter.

Define the sparse supports as

$$\mathbb{S}_\gamma(v) = \text{support}(\gamma_0(v)) = \{j \in \{1, \dots, p\} : |\gamma_{0j}(v)| > 0\},$$

having at most $s^\gamma \leq n / \log(n(p+1))$ nonzero components respectively for all $v \in \mathcal{U}$.

- **Hard-thresholding**: Set coefficients below threshold to zero.

Define the hard-thresholded estimators as

$$\hat{\gamma}_j(v) = \tilde{\gamma}_j(v) 1\{|\tilde{\gamma}_j(v)| > \varepsilon^\gamma\},$$

for some $\varepsilon^\gamma \geq 0$, where $j = 1, \dots, p$ and $v \in \mathcal{U}$.

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Estimation

Second Step (outcome equation): estimate parameters $\beta_0(u)$

In model 1, we have

$$\begin{aligned} Y_1^* &= X' \beta_0(U), \\ Y_1 &= DY_1^* \end{aligned}$$

We proposed $\tilde{b}(\tau, \nu)$ by solving

$$\min_{b \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n \rho_\tau(Y_{1i} - X_i' b) 1\{Y_{2i} > X_i' \tilde{\gamma}(\nu) > 0\} + \frac{\lambda}{n} \|b\|_{1,n}, \quad (7)$$

for any $\tau, \nu \in \mathcal{U}$.

Define the hard-thresholded estimators as

$$\hat{b}_j(\tau, \nu) = \tilde{b}_j(\tau, \nu) 1\{|\tilde{b}_j(\tau, \nu)| > \varepsilon^b\},$$

for some $\varepsilon^b \geq 0$ where $j = 1, \dots, p$ and $\tau, \nu \in \mathcal{U}$.

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Asymptotic Results

Theorem (Convergence Rate)

Under general conditions, the ℓ_1 -penalized quantile regression in the first two steps is uniformly consistent at the near-oracle rate $\sqrt{\frac{s^i \log((p+1)n)}{n}}$, where $i \in \{\gamma, b\}$.

Key conditions:

- Sparsity:

$$\sup_{v \in \mathcal{U}} \|\gamma(v)\|_0 \leq s^\gamma \quad \text{and} \quad \sup_{u \in \mathcal{U}} \|\beta(u)\|_0 \leq s^b$$

- Restricted eigenvalue:

$$\inf_{v \in \mathcal{U}} \inf_{\substack{\Delta_\gamma \neq \mathbf{0} \\ \Delta_\gamma \in \mathcal{A}_v}} \frac{\Delta_\gamma' E(1\{X_i \gamma_0(v) > \delta_0\} X_i X_i') \Delta_\gamma}{\|\Delta_\gamma\|^2} = \sigma_{\min}^2 > 0$$

$$\inf_{v \in \mathcal{U}} \inf_{\substack{\Delta_b \neq \mathbf{0} \\ \Delta_b \in \mathcal{B}_\tau}} \frac{\Delta_b' E(1\{Y_{2i} > X_i' \gamma_0(v) > 0\} X_i X_i') \Delta_b}{\|\Delta_b\|^2} = \delta_{\min}^2 > 0$$

for all i .

Asymptotic Results

Theorem (Hard-thresholding)

Let $r^\gamma = \sup_{v \in \mathcal{U}} \|\tilde{\gamma}(v) - \gamma(v)\|$. Assume $\inf_{v \in \mathcal{U}} \min_{j \in \mathbb{S}_\gamma(v)} |\gamma_j(v)| > c^\gamma$, where $r^\gamma < c^\gamma$, then

$$\mathbb{S}_\gamma(v) := \text{support}(\gamma(v)) \subseteq \tilde{\mathbb{S}}_\gamma(v) := \text{support}(\tilde{\gamma}(v)) \quad \text{for all } v \in \mathcal{U}. \quad (8)$$

Moreover, the hard-thresholded estimator $\hat{\gamma}(v)$, defined for any $\varepsilon^\gamma \geq 0$ by

$$\hat{\gamma}_j(v) = \tilde{\gamma}_j(v) 1\{|\tilde{\gamma}_j(v)| > \varepsilon^\gamma\}, \quad v \in \mathcal{U}, j = 1, \dots, p, \quad (9)$$

provided that ε^γ is chosen such that $r^\gamma < \varepsilon^\gamma < c^\gamma - r^\gamma$, satisfies

$$\text{support}(\hat{\gamma}(v)) = \mathbb{S}_\gamma(v) \quad \text{for all } v \in \mathcal{U}.$$

Asymptotic Results

Theorem

Under general assumptions, then $\hat{\rho}$ is consistent for ρ_0 , and is asymptotically normal:

$$\sqrt{n}(\hat{\rho} - \rho_0) \rightarrow^d N(0, \Sigma_{\rho}).$$

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Monte Carlo simulations

Performance of estimator $\hat{\rho}$ in normal settings

DGP

$$Y_1^* = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \sigma(X) \cdot \Phi^{-1}(U)$$

$$Y_2^* = \gamma_0 + \gamma_1 X_1 + \gamma_2 X_2 + \Phi^{-1}(V)$$

$$D = 1\{Y_2^* > 0\}$$

$$Y_2 = \max(Y_2^*, 0)$$

$$Y_1 = DY_1^*$$

- $X_1 \sim 0.5 \cdot N_1$, $X_2 \sim U[0, 2]$ and N_1 are independent standard normal variables
- $\beta_0 = \beta_1 = \beta_2 = 1, \gamma_0 = 0.2, \gamma_1 = \gamma_2 = 1$
- U and V are uniform on $(0, 1)$, with their dependence modelled through different copulas, and both are independent of X
- $\sigma(X) = 1$ for the homoscedastic case and $\sigma(X) = 1 + 0.5 \cdot X_1$ for the heteroscedastic case

Monte Carlo simulations

Performance of estimator $\hat{\rho}$ in normal settings

Frank copula V.S. Gaussian copula (homoscedastic case)

Param	True	Bias	RMSE	SD
Censoring $\approx 17\%$				
τ	0.7000	$n = 300$ -0.0212	0.0480	0.0430
τ	0.7000	$n = 600$ -0.0138	0.0332	0.0296
τ	-0.7000	$n = 300$ 0.0163	0.0447	0.0420
τ	-0.7000	$n = 600$ 0.0123	0.0332	0.0302

Frank copula

Param	True	Bias	RMSE	SD
Censoring $\approx 17\%$				
ρ	0.7000	$n = 300$ -0.0291	0.0768	0.0710
ρ	0.7000	$n = 600$ -0.0198	0.0510	0.0466
ρ	-0.7000	$n = 300$ 0.0088	0.0707	0.0702
ρ	-0.7000	$n = 600$ 0.0101	0.0469	0.0462

Gaussian copula

Monte Carlo simulations

Performance of estimator $\hat{\rho}$ in normal settings

Frank copula V.S. Gaussian copula (heteroscedastic case)

Param	True	Bias	RMSE	SD
Censoring $\approx 17\%$				
τ	0.7000	$n = 300$ -0.0225	0.0480	0.0429
τ	0.7000	$n = 600$ -0.0144	0.0332	0.0303
τ	-0.7000	$n = 300$ 0.0181	0.0458	0.0427
τ	-0.7000	$n = 600$ 0.0129	0.0332	0.0307

Frank copula

Param	True	Bias	RMSE	SD
Censoring $\approx 17\%$				
ρ	0.7000	$n = 300$ -0.0287	0.0768	0.0712
ρ	0.7000	$n = 600$ -0.0211	0.0520	0.0474
ρ	-0.7000	$n = 300$ 0.0096	0.0721	0.0718
ρ	-0.7000	$n = 600$ 0.0109	0.0479	0.0463

Gaussian copula

Monte Carlo simulations

Performance of estimator $\hat{\rho}$ in normal settings

Frank Copula: Varying dependence strengths

Param	True	Bias	RMSE	SD
Censoring Rate $\approx 17\%$				
τ	0.3000	$n = 300$ -0.0215	0.0900	0.0875
		$n = 600$ -0.0108		
τ	0.5000	$n = 300$ -0.0202	0.0678	0.0651
		$n = 600$ -0.0165		
τ	0.7000	$n = 300$ -0.0212	0.0480	0.0430
		$n = 600$ -0.0138		
τ	0.9000	$n = 300$ -0.0214	0.0265	0.0160
		$n = 600$ -0.0154		

Monte Carlo simulations

Performance of all estimators in high-dimensional settings

DGP

$$Y_1^* = 1 + X_1 + X_2 + X_3 + \sigma(X) \cdot \Phi^{-1}(U)$$

$$Y_2^* = 2.5 + X_1 + X_2 + X_3 + Z_1 + \Phi^{-1}(V)$$

$$D = 1\{Y_2^* > 0\}$$

$$Y_2 = \max(Y_2^*, 0)$$

$$Y_1 = DY_1^*$$

- The number of covariates is set to $p = 500$
- All covariates are independently drawn from a standard normal distribution
- The choice of penalty level λ follows Belloni and Chernozhukov (AOS, 2011)

Monte Carlo simulations

Performance of estimator $\hat{\gamma}(\nu)$ in high-dimensional settings

Frank copula (Kendall's $\tau = 0.7$, homoscedastic case)

τ	Param	True	$n = 1000$			$n = 2000$		
			Bias	SD	RMSE	Bias	SD	RMSE
0.30	γ_0	1.9756	-0.0055	0.0510	0.0512	0.0002	0.0353	0.0352
	γ_1	1.0000	-0.0544	0.0537	0.0764	-0.0383	0.0413	0.0563
	γ_2	1.0000	-0.0322	0.0617	0.0696	-0.0265	0.0430	0.0504
	γ_3	1.0000	-0.0308	0.0625	0.0696	-0.0211	0.0470	0.0514
	γ_4	1.0000	-0.0544	0.0562	0.0782	-0.0419	0.0411	0.0586
0.50	γ_0	2.5000	-0.0184	0.0458	0.0493	-0.0083	0.0314	0.0325
	γ_1	1.0000	-0.0481	0.0496	0.0691	-0.0350	0.0353	0.0497
	γ_2	1.0000	-0.0267	0.0574	0.0632	-0.0205	0.0399	0.0448
	γ_3	1.0000	-0.0234	0.0577	0.0622	-0.0188	0.0408	0.0449
	γ_4	1.0000	-0.0482	0.0513	0.0704	-0.0374	0.0404	0.0550
0.70	γ_0	3.0244	-0.0130	0.0438	0.0456	-0.0007	0.0308	0.0307
	γ_1	1.0000	-0.0537	0.0540	0.0762	-0.0408	0.0355	0.0540
	γ_2	1.0000	-0.0346	0.0579	0.0674	-0.0261	0.0393	0.0471
	γ_3	1.0000	-0.0310	0.0567	0.0646	-0.0271	0.0400	0.0483
	γ_4	1.0000	-0.0560	0.0523	0.0766	-0.0417	0.0379	0.0563
0.90	γ_0	3.7816	0.1834	0.0587	0.1925	0.2133	0.0377	0.2166
	γ_1	1.0000	-0.1739	0.0695	0.1872	-0.1669	0.0495	0.1741
	γ_2	1.0000	-0.1545	0.0759	0.1721	-0.1407	0.0553	0.1511
	γ_3	1.0000	-0.1463	0.0713	0.1627	-0.1443	0.0543	0.1541
	γ_4	1.0000	-0.1826	0.0686	0.1950	-0.1681	0.0484	0.1749

Monte Carlo simulations

Performance of estimator $\hat{\beta}(u)$ in high-dimensional settings

Frank copula (Kendall's $\tau = 0.7$, homoscedastic case)

τ	Param	True	$n = 1000$			$n = 2000$		
			Bias	SD	RMSE	Bias	SD	RMSE
0.3	β_1	1.00	-0.0645	0.0783	0.1013	-0.0711	0.0551	0.0899
	β_2	1.00	-0.0727	0.0818	0.1094	-0.0601	0.0589	0.0841
	β_3	1.00	-0.0985	0.0793	0.1264	-0.1044	0.0564	0.1186
0.5	β_1	1.00	-0.0587	0.0756	0.0956	-0.0626	0.0511	0.0808
	β_2	1.00	-0.0659	0.0766	0.1010	-0.0564	0.0568	0.0800
	β_3	1.00	-0.0903	0.0689	0.1135	-0.0938	0.0524	0.1074
0.7	β_1	1.00	-0.0550	0.0813	0.0981	-0.0541	0.0546	0.0768
	β_2	1.00	-0.0537	0.0826	0.0984	-0.0511	0.0595	0.0784
	β_3	1.00	-0.0823	0.0717	0.1091	-0.0816	0.0544	0.0980
0.9	β_1	1.00	-0.0472	0.1060	0.1160	-0.0447	0.0719	0.0845
	β_2	1.00	-0.0433	0.1082	0.1164	-0.0420	0.0779	0.0884
	β_3	1.00	-0.0728	0.0941	0.1189	-0.0675	0.0708	0.0977

Monte Carlo simulations

Performance of estimator $\hat{\beta}(u)$ in high-dimensional settings

Frank copula (Kendall's $\tau = 0.7$, homoscedastic case)

τ	Param	True	$n = 1000$			$n = 2000$		
			Bias	SD	RMSE	Bias	SD	RMSE
	Kendall's τ	0.7000	-0.0351	0.0412	0.0212	-0.0200	0.02646	0.0163
0.3	β_0	0.4756	0.1974	0.0767	0.2118	0.1501	0.0667	0.1642
	β_1	1.0000	-0.0421	0.0743	0.0853	-0.0310	0.0598	0.0673
	β_2	1.0000	-0.0368	0.0787	0.0868	-0.0278	0.0660	0.0715
	β_3	1.0000	-0.0455	0.0743	0.0871	-0.0366	0.0596	0.0699
0.5	β_0	1.0000	0.0702	0.0606	0.0927	0.0477	0.0483	0.0679
	β_1	1.0000	-0.0385	0.0580	0.0695	-0.0286	0.0462	0.0542
	β_2	1.0000	-0.0287	0.0625	0.0687	-0.0219	0.0491	0.0537
	β_3	1.0000	-0.0404	0.0552	0.0684	-0.0355	0.0444	0.0568
0.7	β_0	1.5244	0.0232	0.0534	0.0582	0.0195	0.0389	0.0435
	β_1	1.0000	-0.0326	0.0545	0.0635	-0.0267	0.0389	0.0471
	β_2	1.0000	-0.0217	0.0609	0.0646	-0.0166	0.0430	0.0460
	β_3	1.0000	-0.0412	0.0523	0.0665	-0.0327	0.0386	0.0505
0.9	β_0	2.2816	0.0128	0.0606	0.0618	0.0128	0.0422	0.0440
	β_1	1.0000	-0.0259	0.0687	0.0733	-0.0244	0.0481	0.0538
	β_2	1.0000	-0.0253	0.0792	0.0830	-0.0143	0.0524	0.0543
	β_3	1.0000	-0.0315	0.0663	0.0733	-0.0268	0.0492	0.0559

Monte Carlo simulations

Performance of estimator $\hat{\beta}(u)$ in high-dimensional settings

Frank copula (Kendall's $\tau = 0.7$, heteroscedastic case)

τ	Param	True	$n = 1000$			$n = 2000$		
			Bias	SD	RMSE	Bias	SD	RMSE
	Kendall's τ	0.7000	-0.0482	0.0529	0.0223	-0.0316	0.0361	0.0173
0.3	β_0	0.4756	0.1489	0.0903	0.1741	0.1020	0.0833	0.1316
	β_1	0.7378	0.1323	0.0740	0.1515	0.1270	0.0615	0.1411
	β_2	1.0000	-0.0531	0.0896	0.1041	-0.0440	0.0774	0.0889
	β_3	1.0000	-0.0664	0.0840	0.1070	-0.0571	0.0707	0.0908
0.5	β_0	1.0000	0.0410	0.0666	0.0781	0.0233	0.0545	0.0592
	β_1	1.0000	0.0281	0.0600	0.0661	0.0293	0.0466	0.0549
	β_2	1.0000	-0.0385	0.0686	0.0786	-0.0312	0.0533	0.0617
	β_3	1.0000	-0.0517	0.0623	0.0809	-0.0458	0.0462	0.0650
0.7	β_0	1.5244	0.0089	0.0553	0.0560	0.0087	0.0401	0.0409
	β_1	1.2622	-0.0212	0.0560	0.0598	-0.0145	0.0368	0.0395
	β_2	1.0000	-0.0271	0.0616	0.0672	-0.0202	0.0415	0.0461
	β_3	1.0000	-0.0459	0.0542	0.0710	-0.0378	0.0383	0.0538
0.9	β_0	2.2816	0.0097	0.0609	0.0616	0.0091	0.0439	0.0447
	β_1	1.6408	-0.0476	0.0631	0.0790	-0.0401	0.0433	0.0590
	β_2	1.0000	-0.0250	0.0755	0.0794	-0.0150	0.0487	0.0509
	β_3	1.0000	-0.0336	0.0675	0.0753	-0.0283	0.0474	0.0551

Outline

- 1 Introduction
- 2 Estimator
- 3 Asymptotic Results (Technical parts)
- 4 Simulation
- 5 An application: Wage Inequality in the United States

Data

- American Community Survey (ACS) from 2001 to 2023
- a 10% stratified random sample is drawn, where stratification is based on state, race, and education level
- The final sample consists of 2,706,867 observations across the 2001-2023 period.

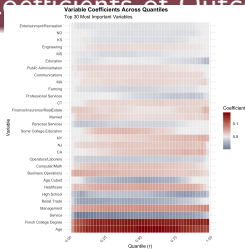
Results

The set of control variables X includes age, age squared, age cubed, three educational attainment dummies, 50 state dummies, a marital status dummy, 8 race category dummies, 14 occupation dummies, and 18 industry dummies.

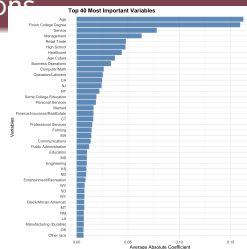
The degree of Selection

Year	Male	Female
2001	-0.164	-0.045
2002	-0.150	-0.060
2003	-0.150	-0.030
2004	-0.135	-0.105
2005	-0.120	-0.030
2006	-0.120	-0.045
2007	-0.105	0.000
2008	-0.105	-0.015
2009	-0.075	-0.015
2010	-0.120	-0.015
2011	-0.075	-0.015
2012	-0.105	0.000
2013	-0.105	-0.015
2014	-0.090	0.000
2015	-0.075	-0.015
2016	-0.090	-0.030
2017	-0.090	-0.030
2018	-0.075	-0.030
2019	-0.090	-0.015
2020	-0.090	-0.045
2021	-0.090	-0.075
2022	-0.090	-0.060
2023	-0.075	-0.045

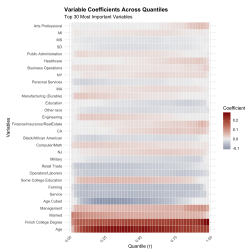
Results: Coefficients of Outcome Equations



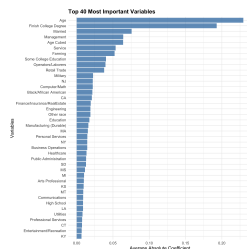
(a) Female: coefficients across quantiles



(b) Female: top 40 important variables



(c) Male: coefficients across quantiles



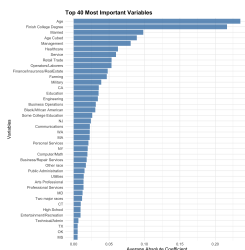
(d) Male: top 40 important variables

Selection-corrected coefficients for the year 2001.

Top 40 Most Important Variables

Variables	Average Absolute Coefficient
Public College Degree	0.20
Age	0.19
Healthcare	0.18
Management	0.17
Service	0.16
Age Cohort	0.15
Recent Travel	0.14
Homes	0.13
CR	0.12
Business Operations	0.11
Education	0.10
Operational Control	0.09
High School	0.08
Complex Work	0.07
Franchise/Import/Export/Date	0.06
Personal Income	0.05
Age	0.04
IRA	0.03
Engineering	0.02
Self-Perceived	0.01
NI	0.00
MS	0.00
Professional Services	0.00
Communications	0.00
EBNA/NI/US Area	0.00
MS	0.00
Some College/Associate	0.00
Overseas/Repair Services	0.00
CR	0.00
Manufacturing/Quintile	0.00
Public Administration	0.00
Other Income	0.00
Other Asian/Pacific Islands	0.00
CR	0.00
Business	0.00
NI	0.00
Military	0.00
Unlabeled	0.00

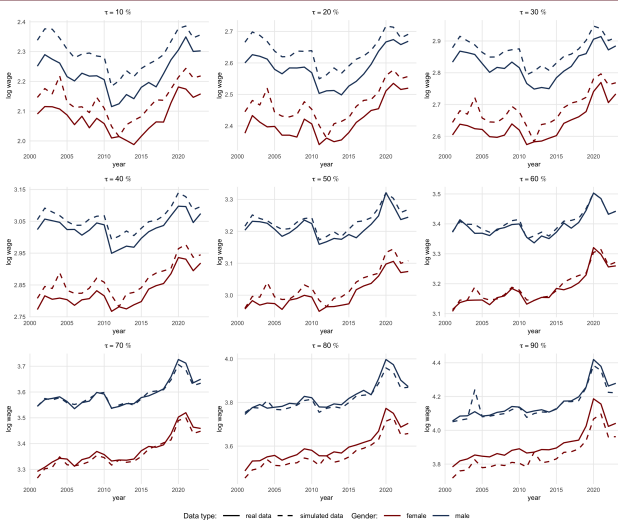
(b) Female: top 40 important variables



(d) Male: top 40 important variables

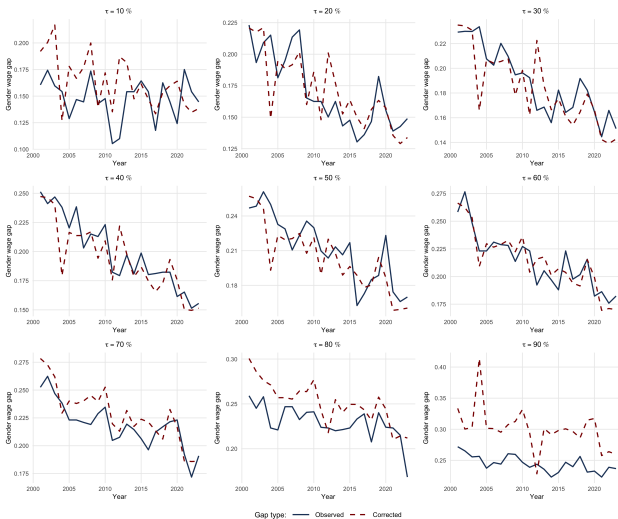
[illegible]

Results: Selection-corrected Wage



Log-wage across quantiles (2001 - 2023)

Results: Wage inequality



Gender wage gaps across quantiles (2001 - 2023)

Thank You!

Questions Welcome

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Job Talk Q&A: Feel free to use the shared Google Doc for questions